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3.0 Summary and Interpretation

2.1 Summarize your findings in a table and provide names for clusters

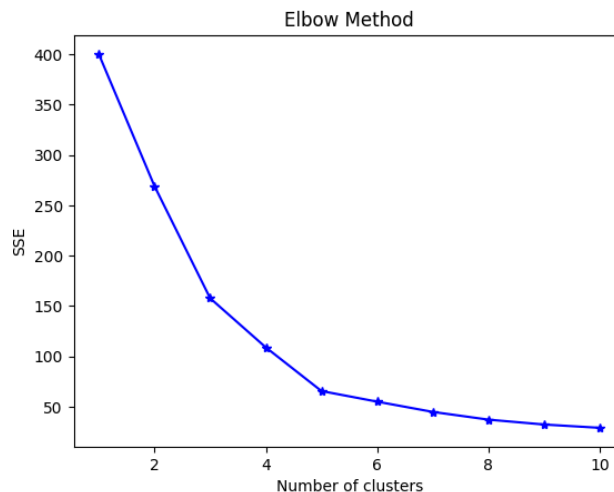
2.2 Provide an interpretation of each cluster

1.0 Summary Overview

In this summary, will discuss more about unsupervised machine learning. There are two algorithms which are k-means and hierarchical. They are different from supervised machine learning and they don't focus on the target.

K-means clustering

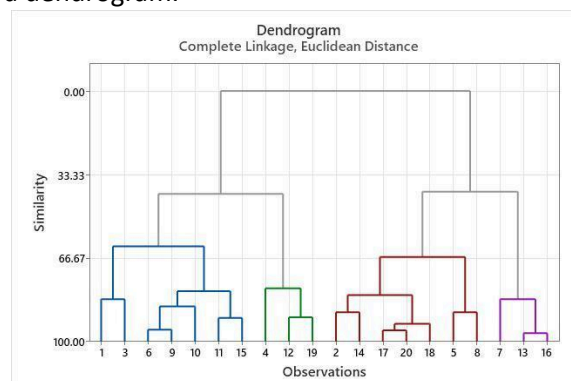
- ❖ The K- denotes the cluster and every cluster there is centroid and how to measure them is by SSE (Sum of Squared Errors) and elbow method.



- The elbow is determined by the graph and the cluster. So on number 5 is the k.

Hierarchical clustering

- ❖ Hierarchical clustering is an unsupervised machine learning technique that groups data into clusters based on similarity. Unlike K-means clustering, it does not require the number of clusters to be specified in advance. Instead, it builds a hierarchy of clusters in the form of a tree structure called a dendrogram.



2.0 Building Clustering Models using Numerical Data

2.1 Data exploration and manipulation on numerical data

Upon looking at the data, there is NaN value which falls under categorical data, Telephone. Numerical data no NaN value.

```
<class 'pandas.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 21 columns):
#   Column                                     Non-Null Count  Dtype
---  -
0   Account Status                            1000 non-null   str
1   Duration in month                        1000 non-null   int64
2   Credit History                            1000 non-null   str
3   Purpose                                  1000 non-null   str
4   Credit amount                            1000 non-null   int64
5   Savings account or bonds                 1000 non-null   str
6   Present employment since                 1000 non-null   str
7   Instalment rate in percentage of disposable income 1000 non-null   int64
8   Personal status and sex                  1000 non-null   str
9   Other debtors or guarantors              1000 non-null   str
10  Present residence since                   1000 non-null   int64
11  Property                                  1000 non-null   str
12  Age in years                             1000 non-null   int64
13  Other installment plans                  1000 non-null   str
14  Housing                                  1000 non-null   str
15  Number of existing credits at this bank  1000 non-null   int64
16  Job                                       1000 non-null   str
17  Number of people being liable to provide maintenance for 1000 non-null   int64
18  Telephone                                404 non-null    str
19  Foreign worker                           1000 non-null   str
20  Target                                   1000 non-null   int64
dtypes: int64(8), str(13)
memory usage: 384.0 KB
```

Table 01: NaN value

The focus was on categorical data but numerical data for clustering methods.

1.1 Data exploration and manipulation ¶

Categorical columns were excluded from the dataset because distance-based clustering models operate on numerical features and are not suitable for raw categorical data.

```
# removing those categorical data as it only uses numerical data
remove_features = ['Account Status', 'Credit History', 'Purpose', 'Savings account or bonds', 'Present employment since',
                  'Personal status and sex', 'Other debtors or guarantors', 'Property', 'Other installment plans', 'Housing',
                  'Job', 'Telephone', 'Foreign worker']

new_data = bank_data.drop(columns = remove_features)
```

Table 02: Categorical data removing

	Duration in month	Credit amount	Instalment rate in percentage of disposable income	Present residence since	Age in years	Number of existing credits at this bank	Number of people being liable to provide maintenance for	Target
0	6	1169	4	4	67	2	1	1
1	48	5951	2	2	22	1	1	2
2	12	2096	2	3	49	1	2	1
3	42	7882	2	4	45	1	2	1
4	24	4870	3	4	53	2	2	2

Table 03: Numerical data after removed categorical data

Now it is ready to build two different clustering models. Target will not be used as this is not supervisor learning. Mainly will manipulate “Duration in month”, “Credit amount”, “Instalment rate in percentage of disposable income”, “Present residence since”, “Age in years”, “Number of existing credits at this bank” or “Number of people being liable to provide maintenance for” for the

clustering result.

2.2 Build TWO different clustering models:

This part where I will build two different clustering algorithms but will perform 4 clustering each model. I will be comparing between k-means and hierarchical, which is the best algorithm and best model of all. And surprisingly, the hierarchical algorithm under linkage has a ward which is default mode, average, single and complete. So I will perform each of each linkage also and the total model will be 20 models.

KMean & Hierarchical

Model Features Clusters value Silhouette Score -> bar chart					
Sample	KMean / Hierarchical	VarA, VarB	Cluster	Silhouette Score	Linkage
Sample	Kmean / Hierarchical	Duration in month, Credit amount	1	0.9	Nil / ward

Table 04: Sample of the model table

2.2.1 K-means Clustering

For k-means clustering, there are 4 models which hold different features inside. There will be 2 to 3 features performing their silhouette score to put one of the best scores. Same as hierarchical features.

Actual	KMean	VarA, VarB, VarC
01	Kmean	"Duration in month", "Credit amount"
02	Kmean	"Age in years", "Number of existing credits at this bank", "Number of people being liable to provide maintenance for"
03	Kmean	"Duration in month", "Credit amount", "Age in years"
04	Kmean	"Age in years", "Number of existing credits at this bank"

Table 05: Features that used for k-means

This will analyze each set of the model and check each clustering and silhouette score.

2.2.2 Hierarchical Clustering

For hierarchical clustering, there are 16 sets of models and hold different features inside inclusive with linkage. Each linkage given will have 4 different sets of models performing their out-most performance. I will use all of it and see each result and see which is the most good performance linkage in hierarchical clustering.

Actual	Hierarchical	VarA, VarB, VarC	Linkage
05	Hierarchical	"Duration in month", "Credit amount"	Ward
06	Hierarchical	"Age in years", "Number of existing credits at this bank", "Number of people being liable to provide maintenance for"	Ward
07	Hierarchical	"Duration in month", "Credit amount", "Age in years"	Ward
08	Hierarchical	"Age in years", "Number of existing credits at this bank"	Ward
Actual	Hierarchical	VarA, VarB, VarC	Linkage
09	Hierarchical	"Duration in month", "Credit amount"	Average
10	Hierarchical	"Age in years", "Number of existing credits at this bank", "Number of people being liable to provide maintenance for"	Average
11	Hierarchical	"Duration in month", "Credit amount", "Age in years"	Average
12	Hierarchical	"Age in years", "Number of existing credits at this bank"	Average
Actual	Hierarchical	VarA, VarB, VarC	Linkage
13	Hierarchical	"Duration in month", "Credit amount"	Single
14	Hierarchical	"Age in years", "Number of existing credits at this bank", "Number of people being liable to provide maintenance for"	Single
15	Hierarchical	"Duration in month", "Credit amount", "Age in years"	Single
16	Hierarchical	"Age in years", "Number of existing credits at this bank"	Single
Actual	Hierarchical	VarA, VarB, VarC	Linkage
17	Hierarchical	"Duration in month", "Credit amount"	Complete
18	Hierarchical	"Age in years", "Number of existing credits at this bank", "Number of people being liable to provide maintenance for"	Complete
19	Hierarchical	"Duration in month", "Credit amount", "Age in years"	Complete
20	Hierarchical	"Age in years", "Number of existing credits at this bank"	Complete

Table 06: Features that used for hierarchical

2.3 Clustering Analysis Using Numerical data

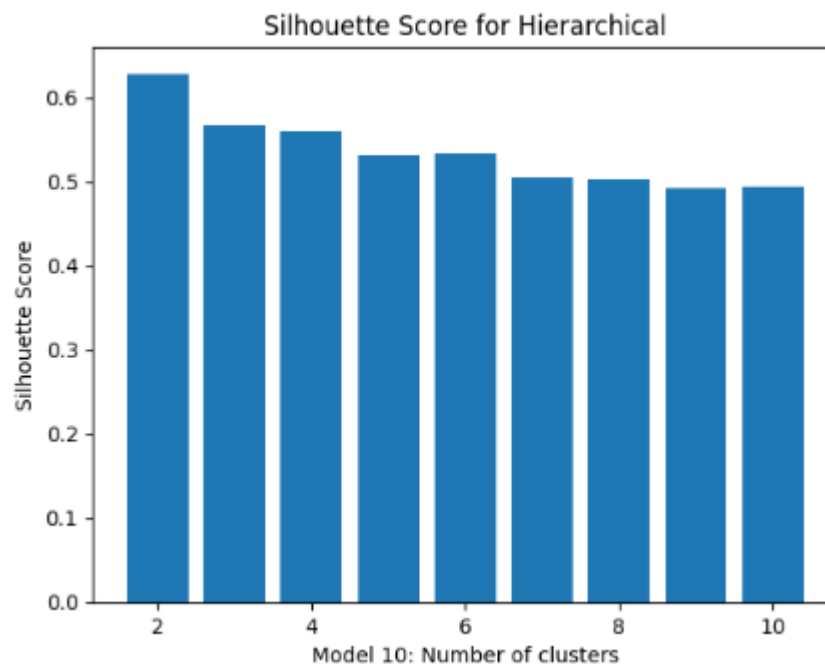


Table 07: Sample table of Silhouette Score and Total Clustering

This is the table shown for model 10 and still has individual model clustering and after this table will also compare each silhouette score for the highest score for the final model selection.

[illegible]

Table 08: either k-mean or hierarchical

This part of clustering is choosing which model is best for the model interpretation. As I had encountered a few models that had more zeros' than 1.

[illegible]

Table 09: Sample of more zeros' than 1

2.4 Evaluate and Compare the Models

Final Result of K-Mean & Hierarchical

Actual	KMean	VarA, VarB, VarC	Cluster	Silhouette Score	
01	Kmean	"Duration in month", "Credit amount"	2	0.5801912097451697	
02	Kmean	"Age in years", "Number of existing credits at this bank", "Number of people being liable to provide maintenance for"	6	0.6002073234036293	
03	Kmean	"Duration in month", "Credit amount", "Age in years"	2	0.4346161965920348	
04	Kmean	"Age in years", "Number of existing credits at this bank"	8	0.5862412201715027	
Actual	Hierarchical	VarA, VarB, VarC	Cluster	Silhouette Score	Linkage
05	Hierarchical	"Duration in month", "Credit amount"	2	0.5389876442823875	Ward
06	Hierarchical	"Age in years", "Number of existing credits at this bank", "Number of people being liable to provide maintenance for"	6	0.600908986378681	Ward
07	Hierarchical	"Duration in month", "Credit amount", "Age in years"	2	0.4606601589330235	Ward
08	Hierarchical	"Age in years", "Number of existing credits at this bank"	5	0.5881274127749511	Ward
Actual	Hierarchical	VarA, VarB, VarC	Cluster	Silhouette Score	Linkage
09	Hierarchical	"Duration in month", "Credit amount"	2	0.6217746575789506	Average
10	Hierarchical	"Age in years", "Number of existing credits at this bank", "Number of people being liable to provide maintenance for"	2	0.628619413638489	Average
11	Hierarchical	"Duration in month", "Credit amount", "Age in years"	2	0.7466770895205375	Average
12	Hierarchical	"Age in years", "Number of existing credits at this bank"	2	0.6233747709090849	Average
Actual	Hierarchical	VarA, VarB, VarC	Cluster	Silhouette Score	Linkage
13	Hierarchical	"Duration in month", "Credit amount"	2	0.782247185921756	Single
14	Hierarchical	"Age in years", "Number of existing credits at this bank", "Number of people being liable to provide maintenance for"	2	0.6227829518889962	Single
15	Hierarchical	"Duration in month", "Credit amount", "Age in years"	2	0.7822310431543447	Single
16	Hierarchical	"Age in years", "Number of existing credits at this bank"	2	0.6233747709090849	Single
Actual	Hierarchical	VarA, VarB, VarC	Cluster	Silhouette Score	Linkage
17	Hierarchical	"Duration in month", "Credit amount"	2	0.7466977175431723	Complete
18	Hierarchical	"Age in years", "Number of existing credits at this bank", "Number of people being liable to provide maintenance for"	2	0.6236002058945159	Complete
19	Hierarchical	"Duration in month", "Credit amount", "Age in years"	2	0.7221967415520961	Complete
20	Hierarchical	"Age in years", "Number of existing credits at this bank"	2	0.6214076596358324	Complete

Table 10: Sample of the Silhouette Score and Clustering

After done with all the 20 models with k-means and hierarchical, the highest silhouette score are:

1. Model 13
2. Model 15
3. Model 17
4. Model 11
5. Model 19
6. Model 10

This part will explain in detail as above mentioned earlier there are clusters that have more zeros' than 1. Even though the top model has the highest silhouette score, there are clusters that don't work quite well.

Model that cluster 0 have value and cluster 1 only 1 number**



Model 13 & 15

Hierarchical -> Linkage = Single

Feature user: "Duration in month", "Credit amount".

Actual	Hierarchical	VarA, VarB, VarC	Cluster	Silhouette Score	Linkage
13	Hierarchical	"Duration in month", "Credit amount"	2	0.782247185921756	Single
15	Hierarchical	"Duration in month", "Credit amount", "Age in years"	2	0.7822310431543447	Single

Model 13, 15 result:

- Cluster 0 ≈ almost all observations
- Cluster 1 = ONE outlier

Model that work out well:

Model 11 & 17

Actual	Hierarchical	VarA, VarB, VarC	Cluster	Silhouette Score	Linkage
17	Hierarchical	"Duration in month", "Credit amount"	2	0.7466977175431723	Complete
11	Hierarchical	"Duration in month", "Credit amount", "Age in years"	2	0.7466770895205375	Average
19	Hierarchical	"Duration in month", "Credit amount", "Age in years"	2	0.7221967415520961	Complete
10	Hierarchical	"Age in years", "Number of existing credits at this bank", "Number of people being liable to provide maintenance for"	2	0.628619413638489	Average

Model 11, 17 result:

- Cluster 0 has 987 observations
- Cluster 1 has 13 observations

Model 19 result:

- Cluster 0 has 15.5 observations
- Cluster 1 has 84.5 observations

Model 10 result:

- Cluster 0 has 23.3 observations
- Cluster 1 has 76.7 observations

Table 11: Table of the top result

The result I took is model 13 first and didn't realise the clustering result.

Model 15 also has the same result as model 13 and is hierarchical and linkage single.

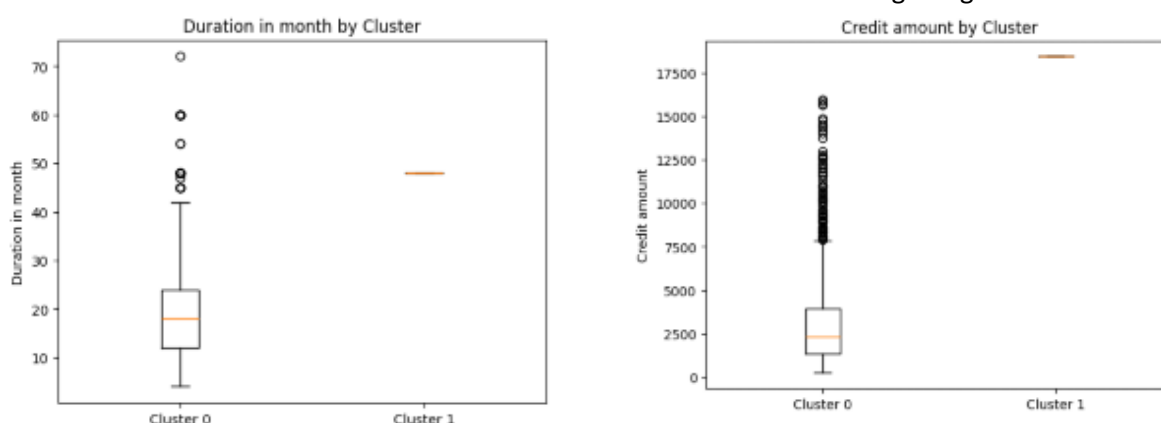


Table 12: Sample of the Model 13 & 15

This is because both models having the same cluster 1 result in the table 11, model 13 and 15 have more zeros' than 1 and cluster 1, one is the outlier. Model 15 I added the feature was "Age in year", and still have the same result as model 13. So this is not a suitable model selection.

Afterward, I chose model 17 on top 3 with the highest silhouette score:

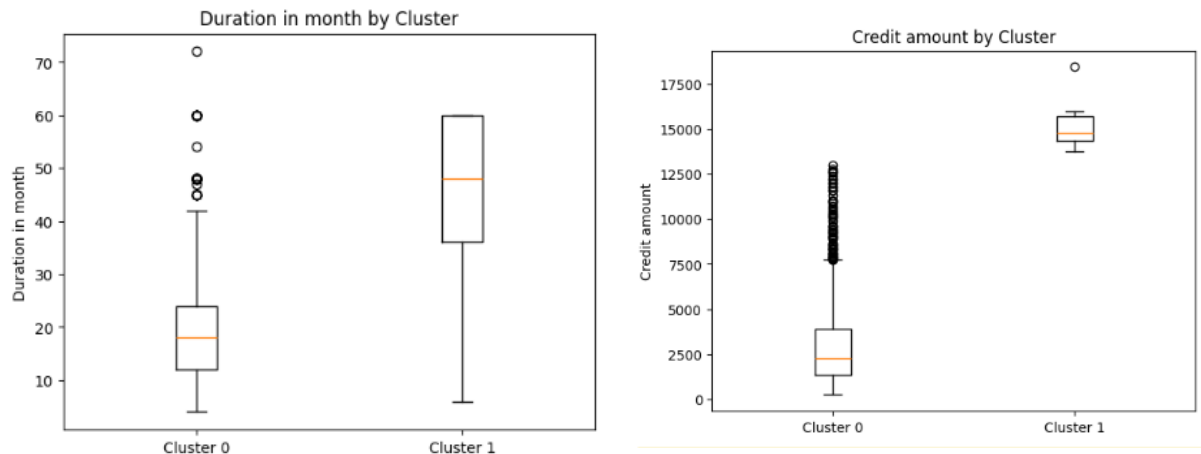


Table 13: Sample of the model 17

As the table is well better than just now model 13 and 15. But there is a part of the cluster that is not well synchronized.

```
Cluster
0    98.7
1     1.3
Name: proportion, dtype: float64
```

Table 14: Model 17 & 11 result of clustering

Cluster 0 has 98.7 while cluster 1 has 1.3. So, this part isn't a good synchronized cluster.

Next, I choose model 19 with the top 5 highest silhouettes score result.

```
Cluster
1    84.5
0    15.5
Name: proportion, dtype: float64
```

Table 15: Model 19 result of clustering

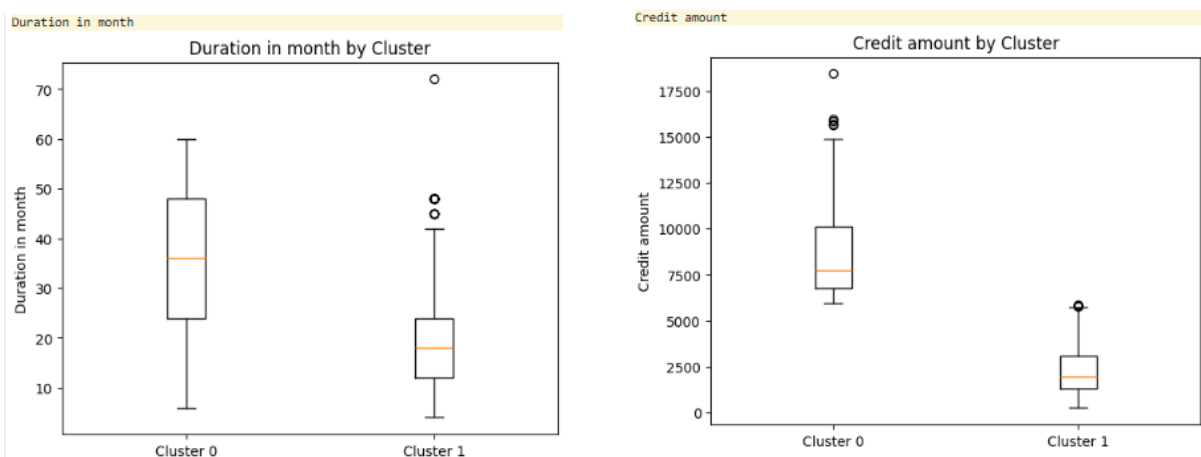


Table 16: Sample of the model 19

This part is better than the above model from top 1 to 4. The silhouette score was 0.7221967415520961. So it is considered the highest still.

Next is model 10:

```
Cluster
1    76.7
0    23.3
Name: proportion, dtype: float64
```

Table 17: Model 10 result of clustering

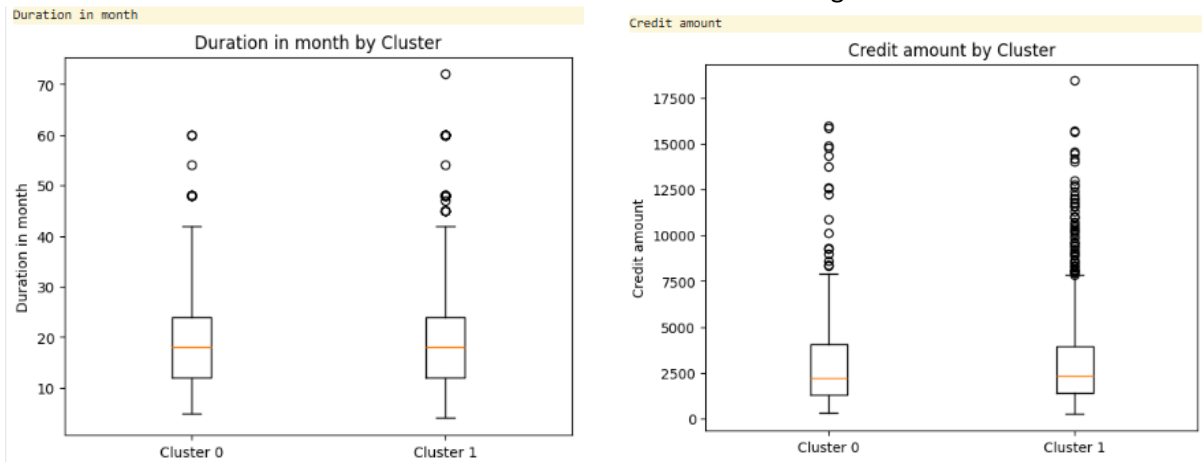


Table 18: Sample of the model 20

Is better than all models but considering the silhouette score of 0.628619413638489.

2.1 Model Comparison Selection

Actual	Hierarchical	VarA, VarB, VarC	Cluster	Silhouette Score	Linkage
10	Hierarchical	"Age in years", "Number of existing credits at this bank", "Number of people being liable to provide maintenance for"	2	0.628619413638489	Average
19	Hierarchical	"Duration in month", "Credit amount", "Age in years"	2	0.7221967415520961	Complete

As it has the highest Silhouette Score and balanced sheet for clustering.

Model 19	Duration in month	Credit amount	Age in years	Number of existing credits at this bank	Number of people being liable to provide maintenance for
cluster 0	Middle	Middle	Low	Low	Low
cluster 1	Low	Low	Low	Low	Low

Model 10	Duration in month	Credit amount	Age in years	Number of existing credits at this bank	Number of people being liable to provide maintenance for
cluster 0	Low	Low	High	Low	Low
cluster 1	Low	Low	Low	Low	Low

Model 19

```
Duration in month , Cluster 0: -> Middle & Cluster 1: <- Low
Credit amount , Cluster 0: -> Middle & Cluster 1: <- Low
Age in years , Cluster 0: -> Low & Cluster 1: <- Low
Number of existing credits at this bank , Cluster 0: -> Low & Cluster 1: <- Low
Number of people being liable to provide maintenance for , Cluster 0: -> Low & Cluster 1: <- Low
```

Model 10

```
Duration in month , Cluster 0: -> Low & Cluster 1: <- Low
Credit amount , Cluster 0: -> Low & Cluster 1: <- Low
Age in years , Cluster 0: -> High & Cluster 1: <- Low
Number of existing credits at this bank , Cluster 0: -> Low & Cluster 1: <- Low
Number of people being liable to provide maintenance for , Cluster 0: -> Low & Cluster 1: <- Low
```

Table 19: Sample of Model Comparison

Model 10 is overall good but everything is low except "Age in year" was high. While, model 19 focuses on "Duration in month" and "Credit amount" which were middle except the rest were low.

Most of the top 6 silhouette scores have almost the same feature. While except model 10. As it was: "Age in years", "Number of existing credits at this bank", "Number of people being liable to provide maintenance for".

Model 19: "Duration in month", "Credit amount", "Age in years" and was hierarchical clustering and used linkage of complete.

In this diagram show in Model 19:

- The cluster 0 and cluster 1, 'Age in years' is low and there is no unequal age group in them.
- The Cluster 0, 'Credit amount' is middle while Cluster 1 is low.
- The Cluster 0, 'Duration in month' is middle while Cluster 1 is low.

While, in the diagram show model 10:

- The cluster 0, 'Age in years' is high while Cluster 1 is low.
- The cluster 0 and cluster 1, 'Credit amount' both are low.
- The cluster 0 and cluster 1, 'Duration in month' both are low.

Table 20: Sample of the comparison

Final Model Selection was Model 19.

Account Status				Cluster	count	Purpose				Cluster	count	Present employment since				Cluster	count
0	No Checking account			0	53	0	business			0	23	0	< 1 year			0	21
1	No Checking account			1	341	1	business			1	74	1	< 1 year			1	151
2	Salary assignments for at least 1 year			0	42	2	car(new)			0	34	2	1 - 4 years			0	42
3	Salary assignments for at least 1 year			1	232	3	car(new)			1	200	3	1 - 4 years			1	297
4	Salary assignments for at least 2 year			0	59	4	car(used)			0	41	4	4- 7 years			0	35
5	Salary assignments for at least 2 year			1	210	5	car(used)			1	62	5	4- 7 years			1	139
6	Salary assignments for at least 3 year			0	1	6	domestic appliances			1	12	6	> 7 years			0	40
7	Salary assignments for at least 3 year			1	62	7	education			0	11	7	> 7 years			1	213
						8	education			1	39	8	unemployed			0	17
						9	furniture/equipment			0	16	9	unemployed			1	45
						10	furniture/equipment			1	165						
						11	others			0	7						
						12	others			1	5						
						13	radio/television			0	21						
						14	radio/television			1	259						
						15	repairs			0	2						
						16	repairs			1	20						
						17	retraining			1	9						
						Savings account or bonds						Cluster	count				
						0	Savings account			0	155						
						1	Savings account			1	845						
Credit History				Cluster	count												
0	all credits at this bank paid back duly			0	10												
1	all credits at this bank paid back duly			1	39												
2	critical account/ other credits existing (not ...			0	40												
3	critical account/ other credits existing (not ...			1	253												
4	delay in paying off in the past			0	27												
5	delay in paying off in the past			1	61												
6	existing credits paid back duly till now			0	65												
7	existing credits paid back duly till now			1	465												
8	no credits taken/ all credits paid back duly			0	13												
9	no credits taken/ all credits paid back duly			1	27												

Other debtors or guarantors				Cluster	count
0	co-applicant	0	10		
1	co-applicant	1	31		
2	guarantor	0	4		
3	guarantor	1	48		
4	none	0	141		
5	none	1	766		
Personal status and sex				Cluster	count
0	female : divorced/separated/married	0	36		
1	female : divorced/separated/married	1	274		
2	male : divorced/separated	0	9		
3	male : divorced/separated	1	41		
4	male : married/widowed	0	6		
5	male : married/widowed	1	86		
6	male : single	0	104		
7	male : single	1	444		
Other installment plans				Cluster	count
0	bank	0	27		
1	bank	1	112		
2	none	0	119		
3	none	1	695		
4	stores	0	9		
5	stores	1	38		
Property				Cluster	count
0	building society savings agreement/ life insur...	0	31		
1	building society savings agreement/ life insur...	1	201		
2	car or other, not in attribute 6	0	55		
3	car or other, not in attribute 6	1	277		
4	real estate	0	14		
5	real estate	1	268		
6	unknown / no property	0	55		
7	unknown / no property	1	99		
Housing				Cluster	count
0	for free	0	38		
1	for free	1	70		
2	own	0	91		
3	own	1	622		
4	rent	0	26		
5	rent	1	153		
Number of existing credits at this bank				Cluster	count
0		1	0	90	
1		1	1	543	
2		2	0	61	
3		2	1	272	
4		3	0	3	
5		3	1	25	
6		4	0	1	
7		4	1	5	
Telephone				Cluster	count
0	yes, registered under the customers name	0	102		
1	yes, registered under the customers name	1	302		
Job				Cluster	count
0	management/ self-employed/\nhighly qualified e...	0	59		
1	management/ self-employed/\nhighly qualified e...	1	89		
2	skilled employee / official	0	79		
3	skilled employee / official	1	551		
4	unemployed/ unskilled - non-resident	0	2		
5	unemployed/ unskilled - non-resident	1	20		
6	unskilled - resident	0	15		
7	unskilled - resident	1	185		
Foreign worker				Cluster	count
0	no	0	3		
1	no	1	34		
2	yes	0	152		
3	yes	1	811		

Table 21: Sample of the Clustering on Model 19

With the given data and clustering, each divide with the cluster 0 and cluster 1 not really 50 percent but earlier mentioned on table 15.

3.0 Summary and Interpretation

3.1 Summarize your findings in a table and provide names for clusters

```
color_pool = ['#AFEEEE', # Pale Turquoise (Lightest)
              '#7FFFD4', # Aquamarine (Light)
              '#40E0D0', # Turquoise (Standard)
              '#00CED1', # Dark Turquoise (Medium Dark)
              '#008B8B', # Dark Cyan (Darkest)
              '#20B2AA', # Light Sea Green
              '#48D1CC', # Medium Turquoise
              '#00FFFF', # Cyan / Aqua
              '#66CDAA', # Medium Aquamarine
              '#5F9EA0', # Cadet Blue (muted turquoise)
              '#2E8B57', # Sea Green (borderline turquoise)
              '#DC143C', # Crimson
              '#FFD700', # Gold
              '#800080', # Purple
              '#8B4513', # Brown
              '#EE82EE', # Violet
              '#FFD301', # Mustard Yellow
              '#005AD9', # Royal Blue
              '#FF2400', # Scarlet Red
              '#013220', # Dark Green
              '#FFE5E5', # Light Red
              '#FFC0CB', # Pink
              '#FFB6C1', # Light Pink
              '#FF7F7F', # Light Coral Red
              '#FF6347', # Tomato
              '#FF0000', # Red
              '#B22222', # Firebrick
              '#8B0000', # Dark Red
              '#F2E6FF', # Lavender Mist
              '#E6CCFF', # Light Lavender
              '#D8BFD8', # Thistle
              '#DA70D6', # Orchid
              '#BA55D3', # Medium Orchid
              '#9370DB', # Medium Purple
              '#8A2BE2', # Blue Violet
              '#480082'] # Indigo (Dark Purple)
```

For table 23, is a random color for the below bar chart every time when it runs.

Table 23: Random Color for bar chart visualization on clustering

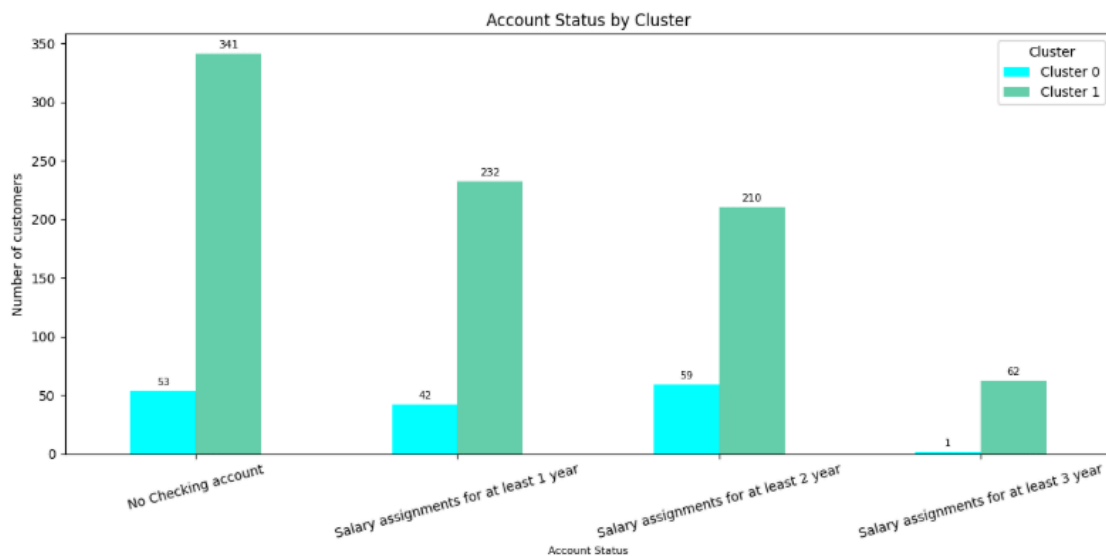


Table 23: Sample of Account Status

In this table, cluster 0 seems to be low compared to cluster 1.

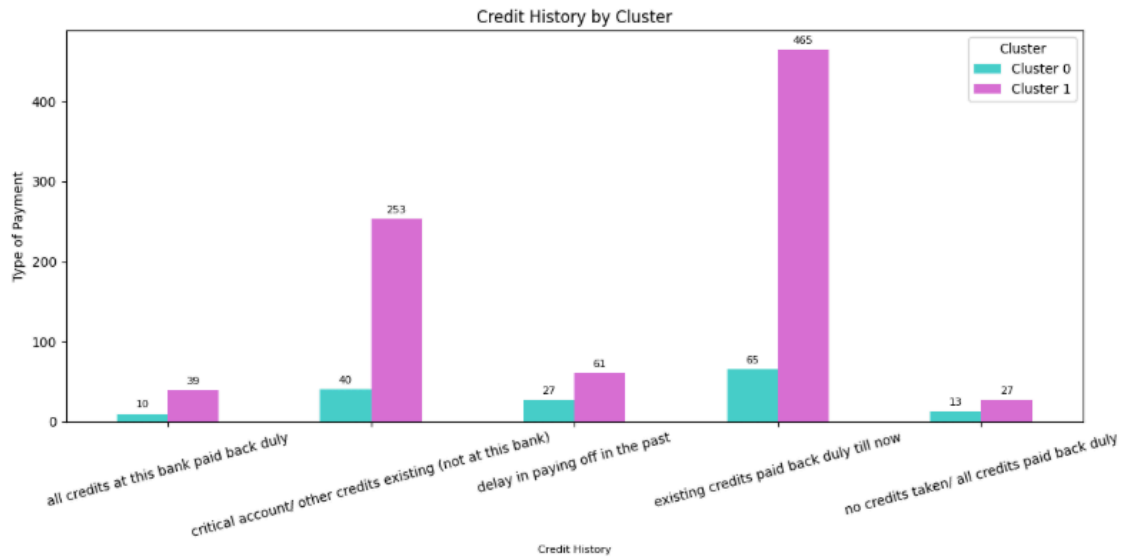


Table 23: Sample of Credit Amount

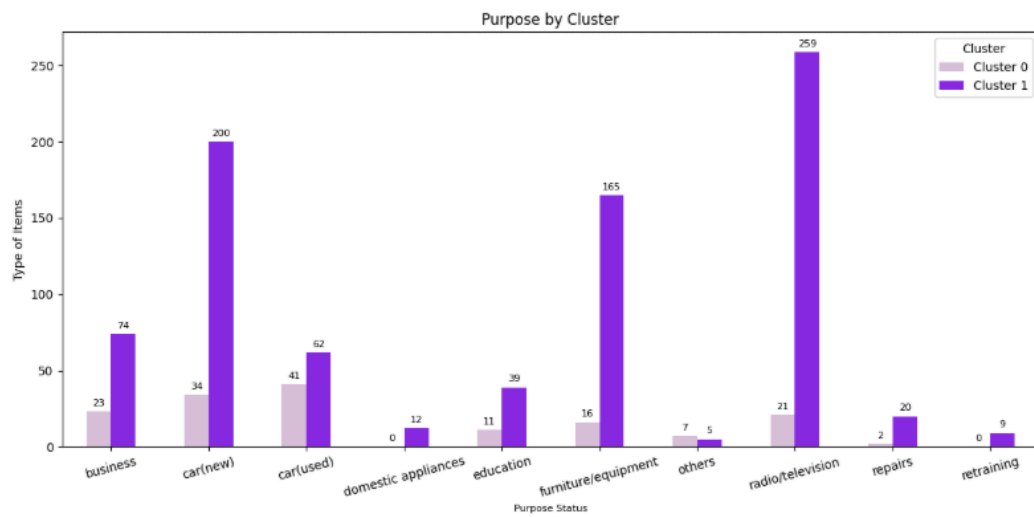


Table 24: Sample of Purpose Status

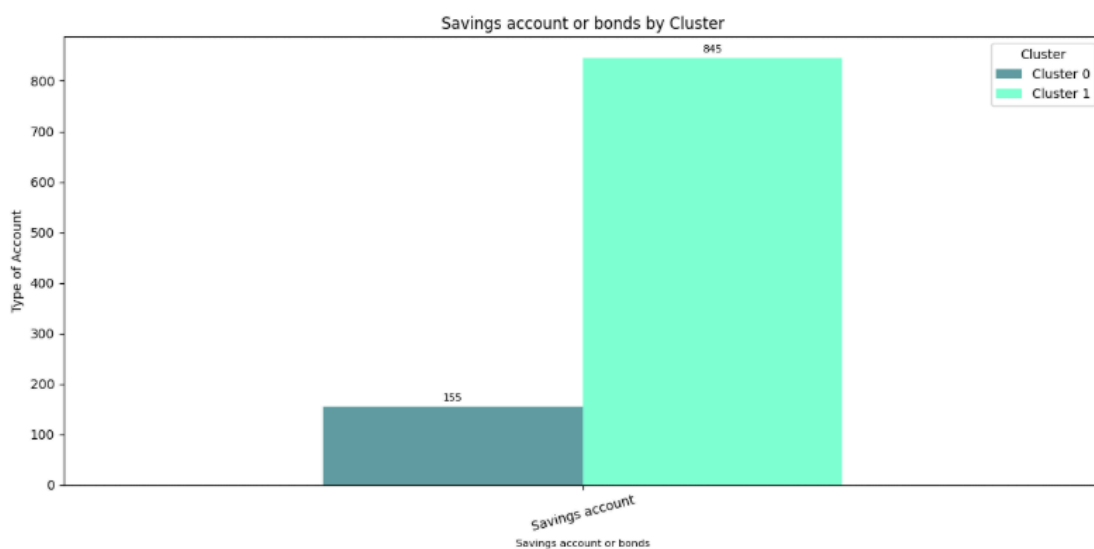


Table 25: Sample of Saving Account

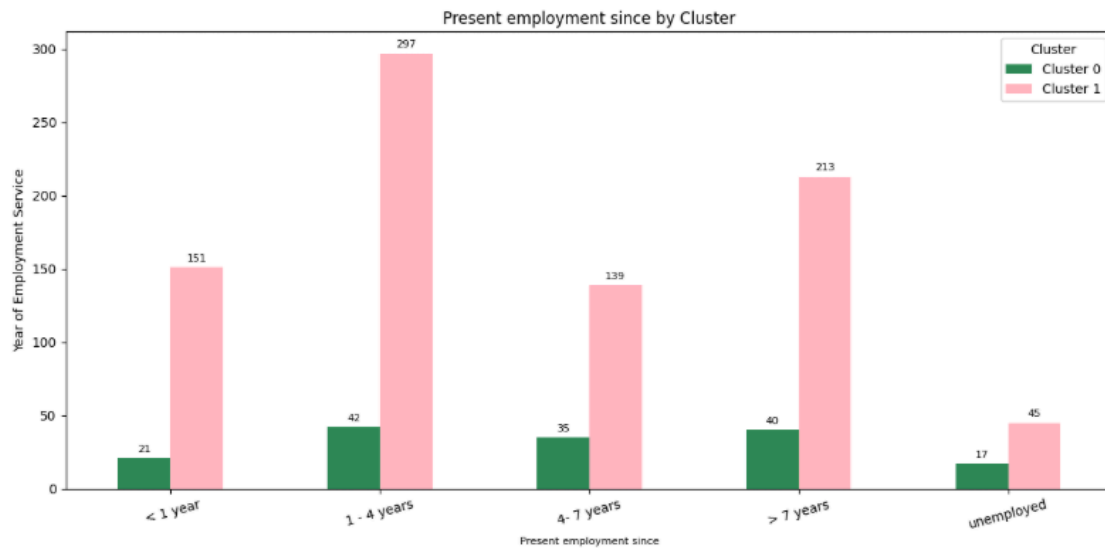


Table 26: Sample of Present employment since

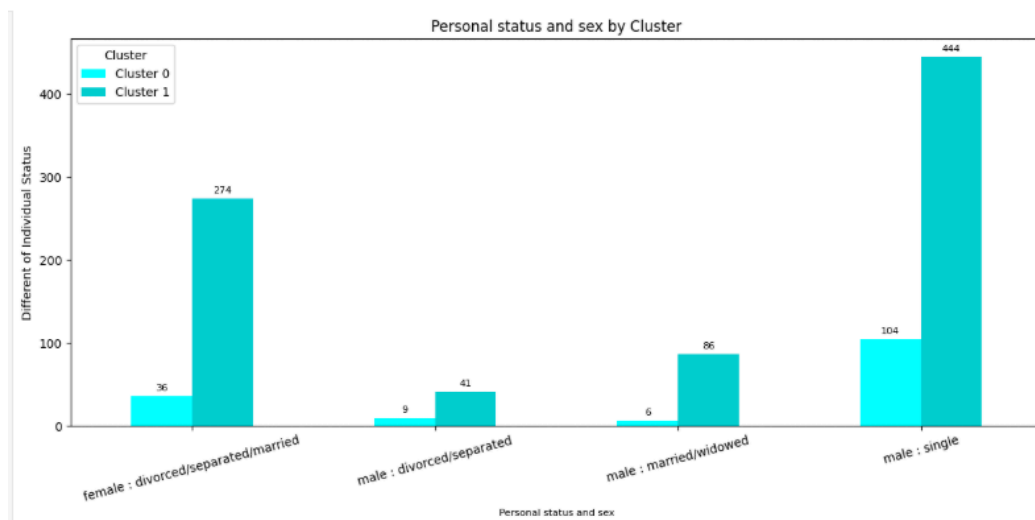


Table 27: Sample of Personal status and sex

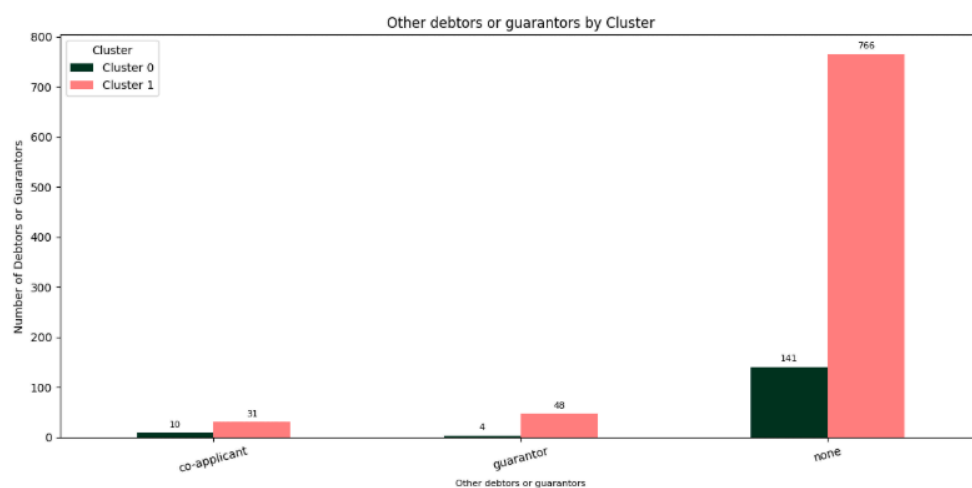


Table 28: Sample of Other debtors or guarantors

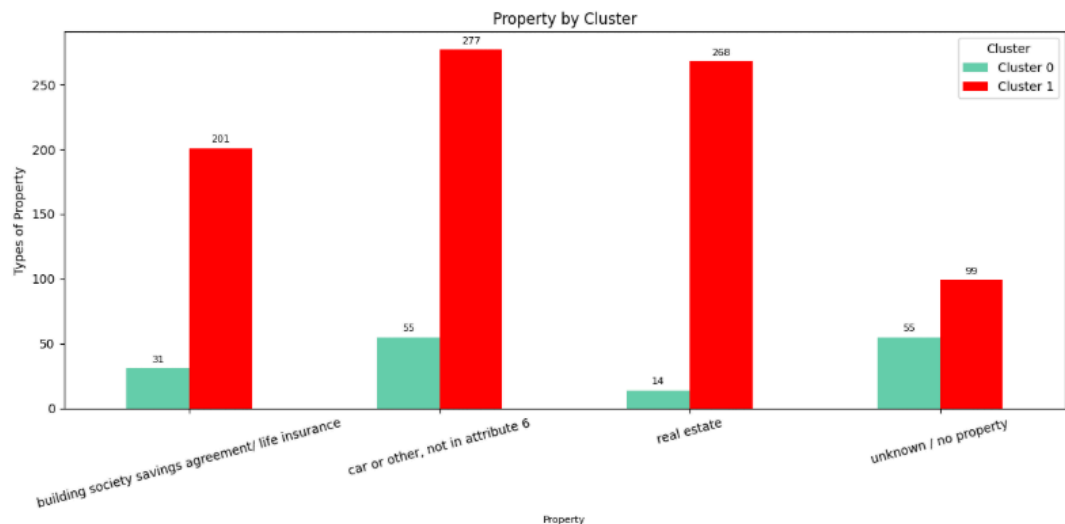


Table 29: Sample of Property

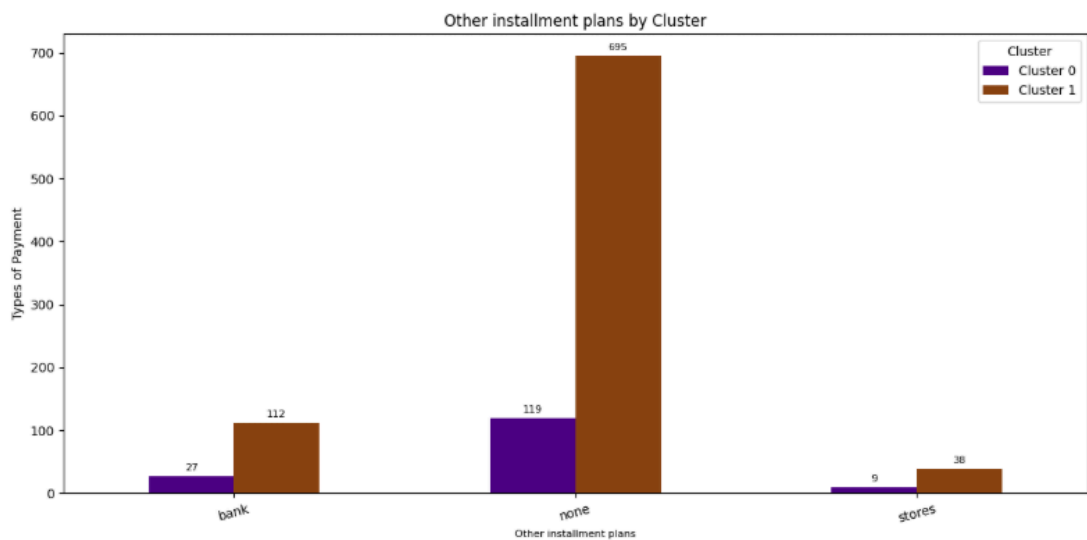


Table 30: Sample of Other Installment Plans

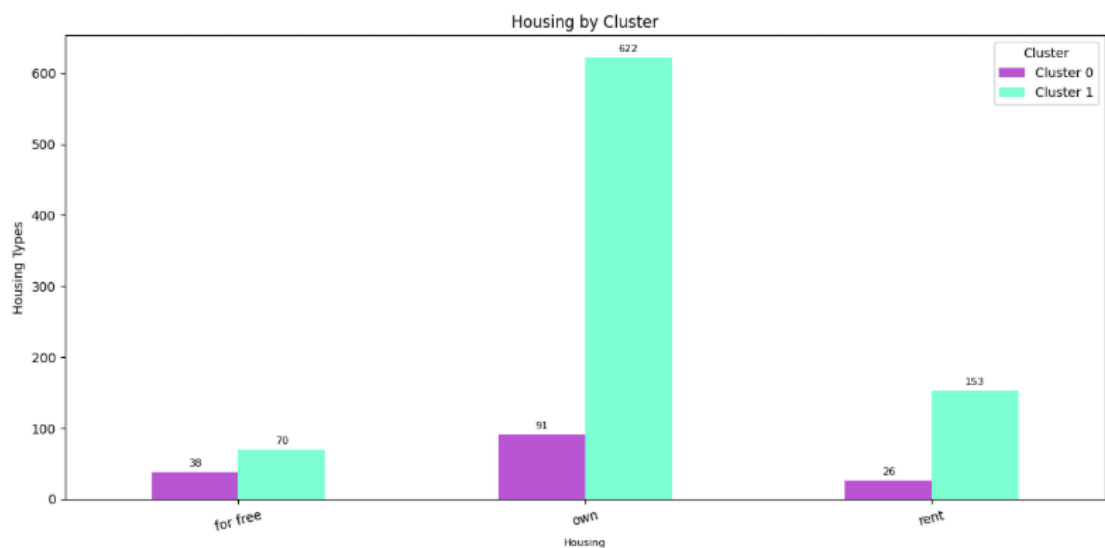


Table 31: Sample of Housing

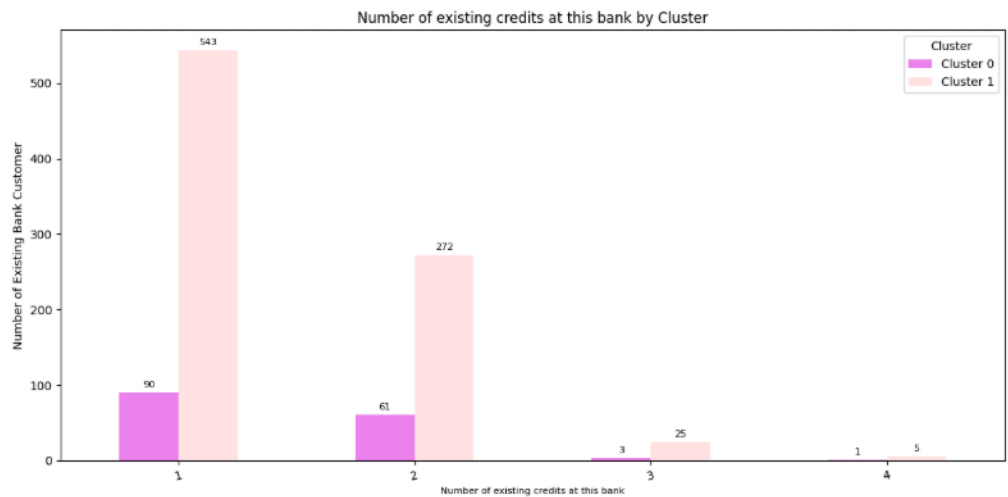


Table 32: Sample of Number of existing credits at this bank

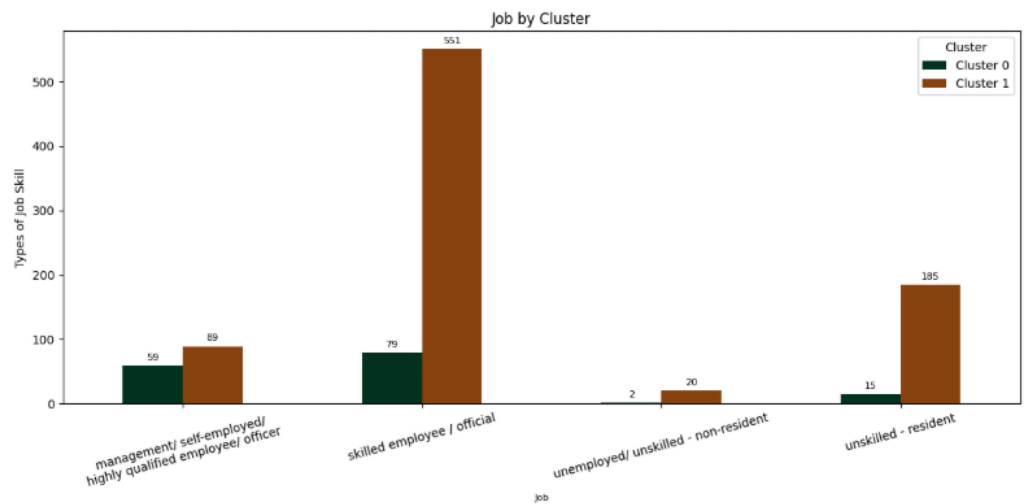


Table 33: Sample of Job

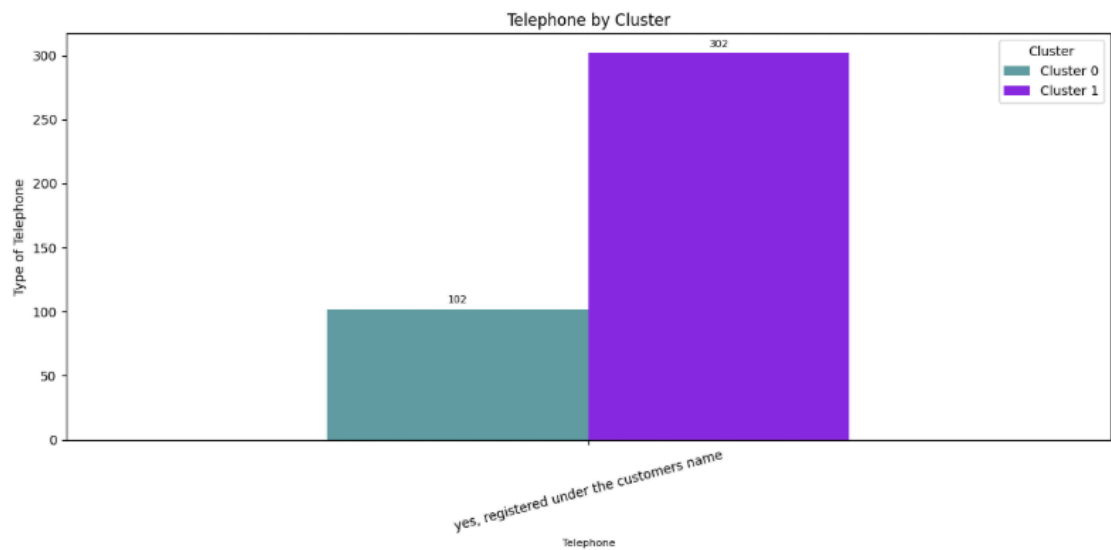


Table 34: Sample of Telephone

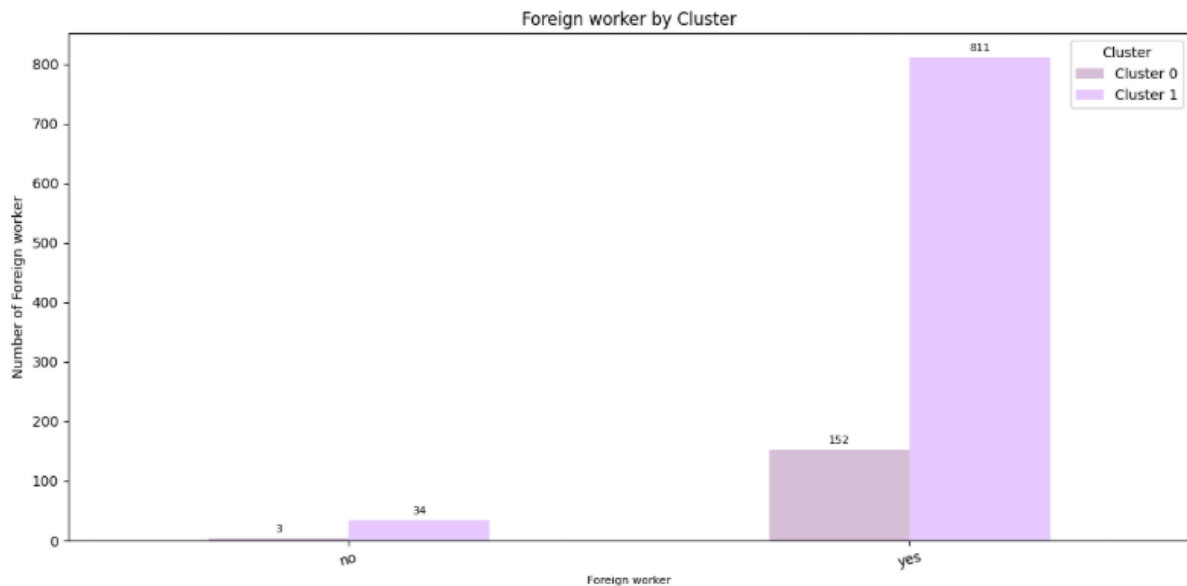


Table 35: Sample of Foreign worker

3.2 Provide an interpretation of each cluster

As for today's presentation, I can make a new way for the barchart as the lecture showed me and explained.

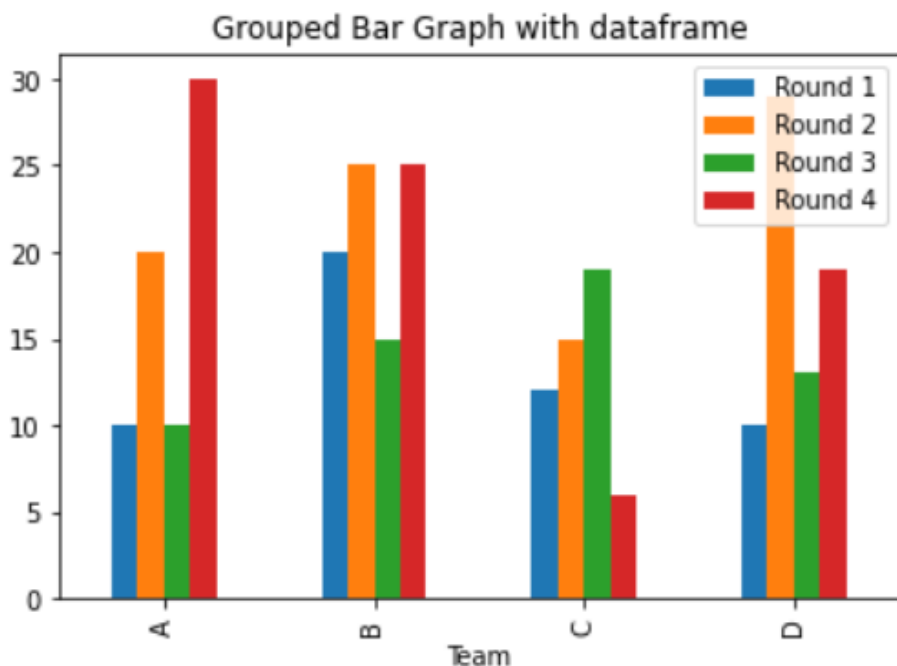


Table 36: Sample of Groupby barchart label by Clustering

In this part of clustering, I can make all the clusters under the name of "Table 29: Sample of Property", in that table on the legend guide. Cluster 0 and cluster 1 which show the value of it and I can split all the cluster 0 and combine it into 1 cluster 0 side-by-side an example on table 36. A will be called cluster 1 and B will be called cluster 1. Then see whether there was any improvement for the business proposal.